Simple Linear Regression

Linearity of Regression

Exercise:

Test $H_0: a_1 = 0$ versus $H_1: a_1 \neq 0$ using t-test at the level of significance $\alpha = 0.05$.

Income x	Food Expenditure y
55	14
83	24
38	13
61	16
33	9
49	15
67	17

Compute the sample correlation coefficient to measure the relationship between X and Y variables. Comment your answer.

Solution:

Linearity of Regression

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Exercise2:

Test $H_0: a_1 = 0$ versus $H_1: a_1 \neq 0$ using the ANOVA approach with significant level, $\alpha = 0.05$

Wheat production	30	28	32	25	25	25	22	24	35	40
Flour price	25	30	27	40	42	40	50	45	30	25

Compute the sample correlation coefficient to measure the relationship between X and Y variables. Comment your answer.

Solution:

Linearity of Regression

Solution: